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To be published in "Small"

February 2018

Center for Functional Nanomaterials
Brookhaven National Laboratory

U.S. Department of Energy
USDOE Office of Science (SC), Basic Energy Sciences (BES) (SC-22)

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Porous carbon-hosted atomically dispersed iron-nitrogen moiety as enhanced electrocatalysts for oxygen reduction reaction in a wide range of pH

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Keywords: Metal-organic frameworks, nonprecious metal nanocatalysts, single-atom catalysts, Fe-N-C electrocatalysts, oxygen reduction reaction

As one of the alternatives to replace precious metal catalysts, transition metal-nitrogen-carbon (M-N-C) electrocatalysts have attracted great research interest due to their low cost and good catalytic activities. Despite nanostructured M-N-C catalysts can achieve good electrochemical performances, they are vulnerable to aggregation and insufficient catalytic sites upon continuous catalytic reaction. Thus, M-N-C's stability and selectivity have not been comparable to their noble metal counterparts. In this work, metal-organic frameworks (MOFs)-derived porous single-atom electrocatalysts (SAEs) were successfully prepared by simple pyrolysis procedure without any further post-treatment. Combining the X-ray absorption near-edge spectroscopy (XANES) and electrochemical measurements, the SAEs have been identified with superior ORR activity and stability compared with Pt/C catalysts in alkaline condition. More impressively, the SAEs also show excellent ORR electrocatalytic performance in both acid and neutral media. This study of nonprecious catalysts provides new

insights on nano-engineering catalytically active sites and porous structures for nonprecious metal ORR catalysis in a wide range of pH.

Oxygen reduction reaction (ORR) plays a crucial role in energy conversion and storage devices, such as fuel cells and metal-air batteries.¹⁻⁵ Nowadays, Pt nanoparticles (NPs) loaded on carbon supports (Pt/C) with the high surface area are considered as the state-of-the-art electrocatalysts for ORR. Whereas, some issues of Pt/C catalysts need to be resolved before the widespread implementation of these devices, including Pt dissolution, catalysts poisoning and poor methanol tolerance.⁶⁻⁸ In addition, the high price and scarcity of Pt also hinder their further commercialization. To address these issues, nonprecious metal nanocatalysts (NPMNs), obtained from earth-abundant elements (Fe, N, C, Co, S and so on), have attracted increasing attention and been studied extensively.⁹⁻¹² Among them, Fe-based carbonaceous catalysts, with Fe-N-C active sites, are the most promising NPMNs for ORR.¹³⁻¹⁵ Their activity and stability in alkaline media can approach or even outperform those of Pt/C.^{16, 17} This has stimulated researchers to develop efficient catalysts with a high density of Fe-N-C active sites through the rational design of their nanostructures. On the other hand, rapidly emerging studies on single-atom catalysts featured with homogeneity of the catalytic active site, low-coordination environment of the metal atom as well as the maximum metal utilization efficiency have received great consideration because of their high electrocatalytic properties.¹⁸⁻²¹ Constructing Fe-N-C structures with single-atom feature is considered as the most efficient strategy, which allows the even dispersion of highly active Fe-N-C active sites and ensures the high ORR performance.^{15, 19, 22-25} For instance, our group reported a facile and universal strategy to synthesize Fe,N-doped carbon nanotube aerogels with single-atom feature, which exhibited excellent electrocatalytic activity and durability for ORR in alkaline condition.¹⁵ Besides, atomically FeN₂ moieties-modified mesoporous carbon,²⁴ N-doped porous carbon²⁵

and cubic carbon frameworks²⁶ with atomically dispersed active sites were reported previously with highly efficient ORR performance in alkaline conditions.

Precisely tuning the morphology of the electrocatalysts to realize large surface area and pore volume is considered as another efficient approach for improving their performances. The porous structures can not only provide more exposed active sites but facilitate mass-transport for electrocatalytic reactions. As typical porous materials, metal-organic frameworks (MOFs), which are composed of well-organized metal centers and organic ligands, are promising templates or precursors for porous nanomaterials in energy applications, including fuel cells, supercapacitors and batteries.²⁷⁻³² In addition to the large surface area, the composition of the resultant M-N-C catalysts can be easily controlled by substituting the metals or linkers. More recently, MOFs-derived M-N-C porous materials have been proven to possess comparable or even better ORR catalytic activity compared with commercial Pt/C.^{12, 33, 34} In comparison with the traditional raw materials, MOFs are ideal precursors, which contain metal, nitrogen and carbon precursors in one single framework.¹² Herein, we developed an efficient strategy to synthesize highly efficient ORR electrocatalysts featured by single active sites and high surface area through simple pyrolysis of MOF precursors. Importantly, these MOFs-derived SAEs display excellent ORR catalytic performance in alkaline, acid and neutral media, which is attributed to the highly exposed homogeneous active sites and morphological advantages. This facile procedure paves a new way for the preparation of SAEs for ORR in a wide range of pH.

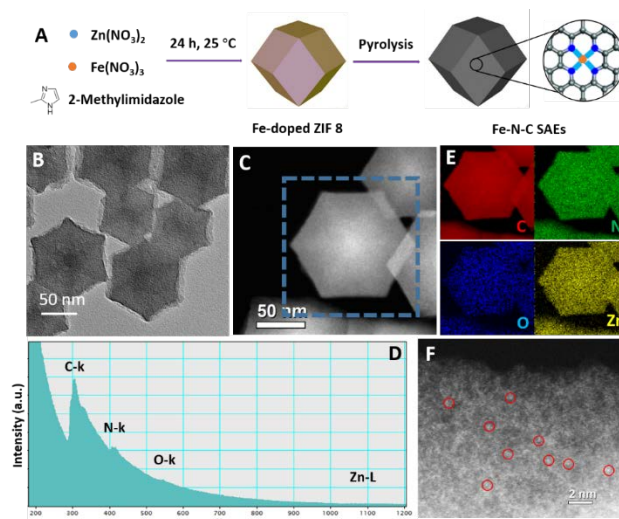


Figure 1. (A) Schematic synthesis of MOF-derived Fe-N-C SAEs. TEM (B), HAADF-STEM (C, F) images of Fe₅₀-N-C-900 SAEs. (D) EELS spectrum of Fe₅₀-N-C-900 SAEs. (E) Elemental distribution of C, N, O, and Zn in Fe₅₀-N-C-900 SAEs.

The preparation of Fe-doped MOF precursors, zeolitic imidazolate frameworks (ZIF-8), were similar to the previous report except for the addition of a certain amount of Fe(NO₃)₃ as shown in Figure 1A.³⁵ The details are available in the experimental section. After the pyrolysis of MOF precursors at different temperatures, a series of Fe_x-N-C-T (x=amount of Fe precursors, T=pyrolysis temperature) SAEs were obtained. Figure 1B shows the typical transmission electron microscopy (TEM) images of Fe₅₀-N-C-900 SAEs. It indicates that the dodecahedral structure was well maintained after pyrolysis. The average size of Fe₅₀-N-C-900 SAEs is around 87 nm. The component of Fe₅₀-N-C-900 SAEs was identified by electron energy loss spectroscopy (EELS) and mapping. The EELS spectrum of Fe₅₀-N-C-900 SAEs in Figure 1D reveals that C, N, O and Zn are the dominant species. From the mapping profile (Figure 1E), we can see that these elements are uniformly distributed throughout the nanostructure. It should be noted that Fe was hard to be detected because of the small amount. The presence of Zn in Fe₅₀-N-C-900 SAEs indicates that Zn was not completely evaporated during the pyrolysis process. The single-atom

features of Fe₅₀-N-C-900 were then confirmed by high-angle annular dark-field scanning transmission electron microscope (HAADF-STEM). Figure 1F clearly shows some bright dots in HAADF-STEM image, ascribing to Zn and Fe atoms, due to the Z-contrast between metal and carbon. These isolated metal atoms confirm the single-atom feature of Fe₅₀-N-C-900 SAEs. To study the effect of Fe concentration and analysis temperature on the morphology, N-C-900 and other SAEs with different Fe were also prepared. The corresponding TEM images in Figure S1 suggest that the size of resultant catalysts increases with the increase of Fe content. Whereas, the annealing temperature does not have much effect on the morphology of Fe-N-C SAEs, as shown in Figure S2.

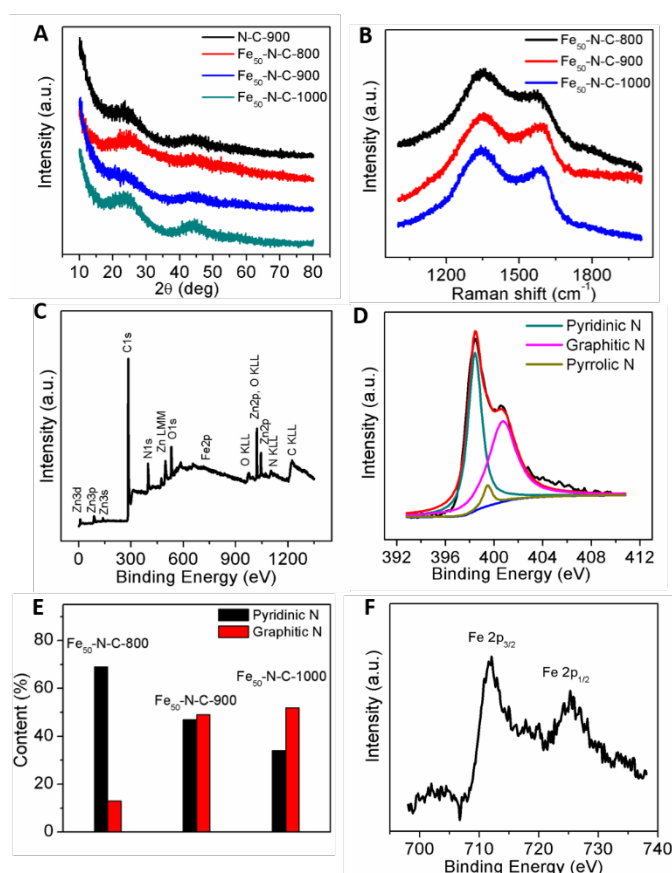


Figure 2. (A) XRD patterns of various catalysts. (B) Raman spectra of Fe₅₀-N-C SAEs with different temperatures. (C) XPS spectrum of Fe₅₀-N-C-900. (D) High-resolution XPS

spectrum of N in Fe₅₀-N-C-900. (E) N content in Fe₅₀-N-C-900. (F) High-resolution XPS spectrum of Fe in Fe₅₀-N-C-900.

X-ray diffraction (XRD) was then carried out to study the crystalline structure of SAEs. Figure 2A shows that two broad peaks at around 24° and 44° were detected in N-C-900 and Fe₅₀-N-C SAEs, corresponding to the (002) and (100) planes of graphitic carbon. The Raman spectra of Fe₅₀-N-C SAEs with different pyrolysis temperature (Figure 2B) show two peaks at Raman shift of ~ 1350 and 1590 cm⁻¹, indicating the D band and G band in graphite, respectively. All Fe₅₀-N-C SAEs show higher D band peak, indicating the large amount of high disordered carbon with abundant defects, which is beneficial for catalytic performance.³⁶⁻³⁸ It should be noted that the content of C, O, N, Zn and Fe in Fe₅₀-N-C-900 SAEs is 80.67, 7.54, 9.79, 1.85 and 0.15 at%, respectively. There is still a small amount of Zn maintained in the final products even under a much higher pyrolysis temperature (~1.8 at% Zn in Fe₅₀-N-C-1000) as shown in Table S1. The information of surface composition and oxidation states were obtained by X-ray photoelectron spectroscopy (XPS) analysis as revealed in Figure 2C. The high-resolution N 1s XPS spectrum in Fe₅₀-N-C-900 (Figure 2D) was further deconvoluted into three peaks at 398.42, 399.45, and 400.69 eV, in correspondence with pyridinic N, pyrrolic N and graphitic N.^{39, 40} Among them, pyridinic N and graphitic N are the dominant species with the atomic percentage of 47.36% and 48.65%, respectively. In comparison with Fe₅₀-N-C-800 and Fe₅₀-N-C-1000, Fe₅₀-N-C SAEs exhibit the highest content of pyridinic N and graphitic N (Figure 2E). These two forms of N were demonstrated to be favorable for ORR.^{15, 41} Figure 2F shows the high-resolution XPS spectrum of Fe 2p in Fe₅₀-N-C SAEs. The two peaks at 711.8 and 725.4 eV indicate the presence of Fe (II) and Fe (III).

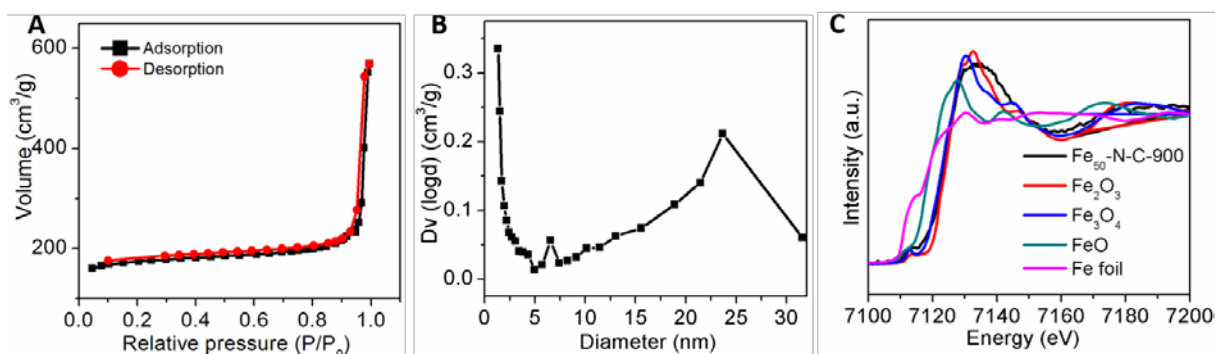


Figure 3. Nitrogen adsorption and desorption isotherms (A) and pore size distribution curve (B) of Fe₅₀-N-C-900. XANES spectra (C) of Fe₅₀-N-C-900.

To get insight into the surface area and porosity property of the MOF-derived SAEs, N₂ adsorption/desorption profile of Fe₅₀-N-C-900 is revealed in Figure 3A and B. The Brunauer-Emmett-Teller surface area and pore volume is 540.3 m²/g and 0.88 cm³/g. The pore size distribution curve indicates that micro and mesopores are dominant in Fe₅₀-N-C-900. The high surface area and large pore volume can contribute faster mass-transport and electron-transfer process, resulting in improved ORR electrocatalysis.² To further confirm the structure of Fe₅₀-N-C-900 SAEs, X-ray absorption analysis was carried out. The X-ray absorption near-edge structure (XANES) in Figure 3C shows that the near-edge absorption energy of Fe₅₀-N-C-900 is between that of standard Fe foil and Fe₂O₃, indicating Fe atoms in the SAEs were positively charged, which was related to the N-coordination in single-atom catalysts.^{22, 39,,}

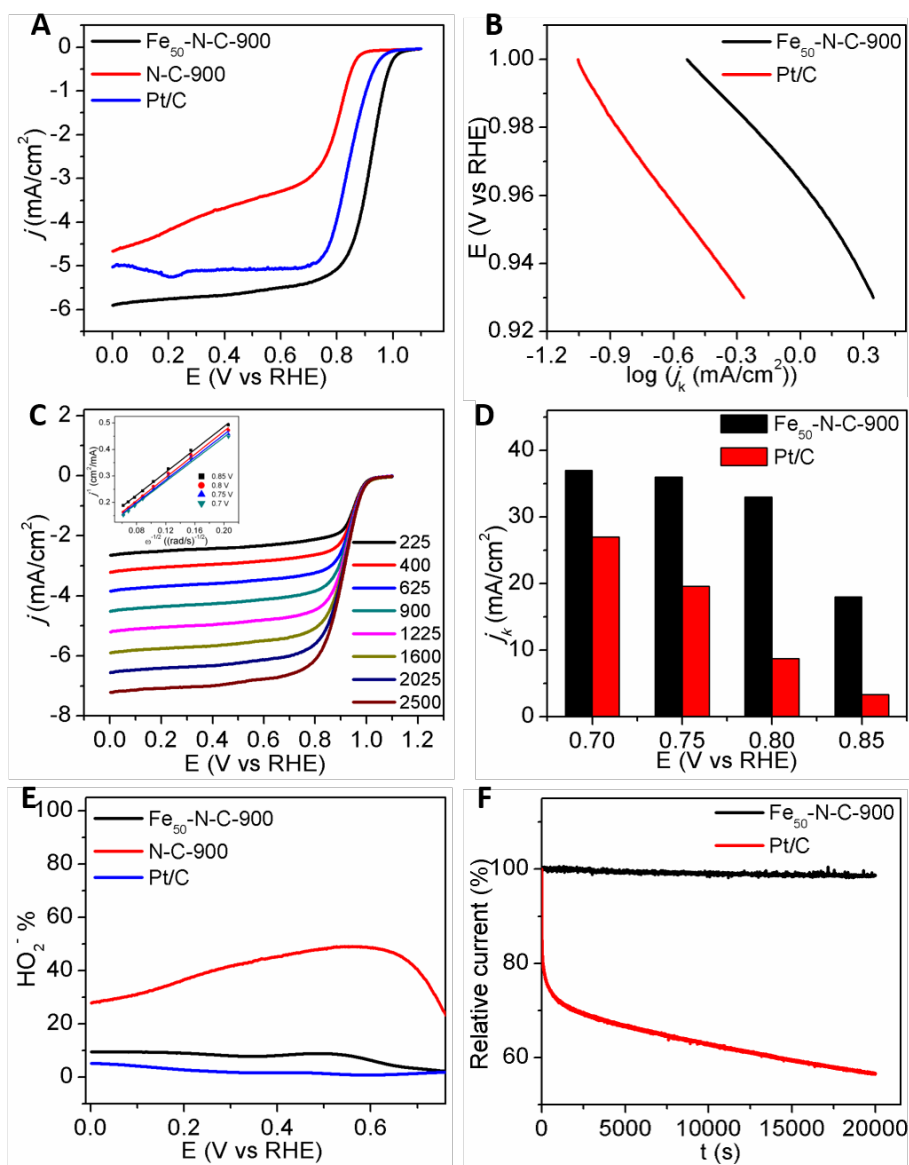


Figure 4. (A) RDE polarization curves of Fe₅₀-N-C-900, N-C-900 and commercial Pt/C catalysts in O₂-saturated 0.1 M KOH solution at a scan rate of 10 mV/s and a rotation rate of 1600 rpm. (B) Tafel plots of Fe₅₀-N-C-900 and Pt/C derived from Figure 3A. (C) RDE polarization curves on Fe₅₀-N-C-900 at different rotation rates and corresponding K-L plots (inset). (D) Kinetic current density (j_k) of the Fe₅₀-N-C-900 and Pt/C at different potentials. (E) HO₂⁻ yield percentage on various catalysts derived from RRDE analysis. (F) Current versus time (i-t) chronoamperometric response of the Fe₅₀-N-C-900 and Pt/C at 0.7 V in O₂-saturated 0.1 M KOH at 200 rpm, respectively. The loading is 0.1 mg/cm² for all catalysts.

Upon the investigation of structural and compositional properties of Fe₅₀-N-C-900 SAEs, we next studied their ORR electrocatalytic performance in an alkaline solution using the rotating disk electrode (RDE). For comparison, the ORR catalytic activity of commercial Pt/C, N-C-900 and Fe-N-C with different synthesis condition were also investigated. Figure 4A shows the RDE polarization curves of C-N-900, Fe₅₀-N-C-900 SAEs and Pt/C catalysts in 0.1 M KOH solution with a scan rate of 10 mV/s and a rotation rate of 1600 rpm. According to the linear sweep voltammetry (LSV) curves, Fe₅₀-N-C-900 SAEs possess the highest ORR activity with a more positive onset potential (1.00 V) and half-wave potential (0.92 V) compared with N-C-900 (0.87 and 0.81V) and Pt/C (0.95 and 0.85 V). Importantly, the performance of Fe₅₀-N-C-900 also outperforms that of other Fe-N-C SAEs obtained under different synthesis condition and the ORR catalysts reported in others' works as shown in Figure S3 and Table S2. To further confirm the structural and compositional advantages of Fe₅₀-N-C-900 SAEs, Tafel plots of Fe₅₀-N-C-900 SAEs and Pt/C were derived from Figure 4A. The Tafel slope (Figure 4B) of Fe₅₀-N-C-900 SAEs is 77 mV/dec, which is smaller than that of Pt/C (86 mV/dec). All these results, including much more positive onset potential and half-wave potential as well as small Tafel slope, suggest that the single active sites are extremely efficient for ORR. Figure 4C provides the RDE polarization curves at rotation rate ranging from 225 to 2500 rpm, which exhibit rotation rate-dependent current density. Based on these LSV curves, Koutecky-Levich (K-L) plots at various potential were obtained and revealed in Figure 3C inset. The good linearity and parallelism indicate the consistent electron-transfer number and first-order reaction kinetics regarding the oxygen concentration. Accordingly, the kinetic current densities (j_k) of Fe₅₀-N-C-900 and Pt/C at certain potentials were calculated. As illustrated in Figure 4D, Fe₅₀-N-C-900 catalysts possess much higher j_k than that of Pt/C. We next conducted the rotating ring-disk electrode (RRDE) tests to clarify the ORR pathway

and HO_2^- yield. Based on the RRDE measurements, the average electron-transfer number on $\text{Fe}_{50}\text{-N-C-900}$ is 3.95, which is similar as that of Pt/C (3.97) as illustrated in Figure S3C. In addition, the HO_2^- yield is less than 10% (Figure 4E), further confirming the four-electron ORR pathway on $\text{Fe}_{50}\text{-N-C-900}$ electrode. Aside from the activity, methanol tolerance is another important factor to evaluate the performance of ORR electrocatalysts in fuel cells. As expected, $\text{Fe}_{50}\text{-N-C-900}$ SAEs present much better tolerance to methanol than Pt/C, as shown in Figure S4. Remarkably, $\text{Fe}_{50}\text{-N-C-900}$ SAEs also possess exceptional long-term durability. In contrast with the substantial current drop-off of Pt/C electrode (Figure 4), $\text{Fe}_{50}\text{-N-C-900}$ exhibited a constant current during the stability test. Additionally, the robustness of the SAEs was further confirmed by the morphology study. No aggregation or collapse was observed in $\text{Fe}_{50}\text{-N-C-900}$ as revealed in Figure S5.

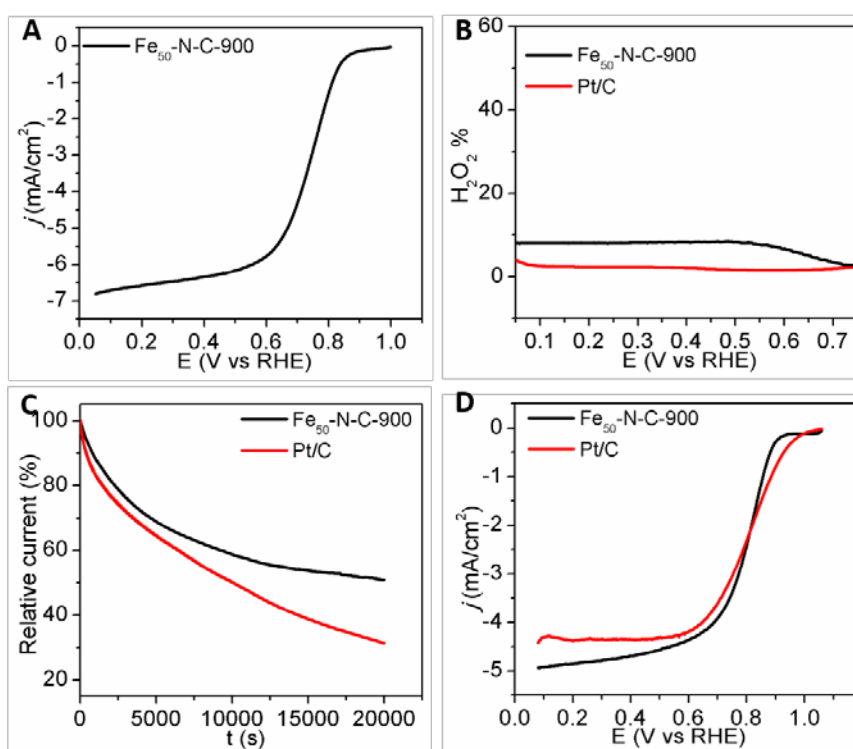


Figure 5. (A) RDE polarization curves of $\text{Fe}_{50}\text{-N-C-900}$ (loading: 0.3 mg/cm^2) in O_2 -saturated 0.1 M HClO_4 solution. (B) The H_2O_2 yield percentage of $\text{Fe}_{50}\text{-N-C-900}$ and Pt/C derived from RRDE analysis. (C) Long-term stability test of $\text{Fe}_{50}\text{-N-C-900}$ and Pt/C in O_2 -saturated 0.1 M

HClO₄ solution at 0.7 V with a rotation rate of 200 rpm. (D) RDE polarization curves of Fe₅₀-N-C-900 (loading: 0.1 mg/cm²) and Pt/C (loading: 0.1 mg/cm²) in O₂-saturated 0.1 M PBS solution (pH=7.4).

The exceptional ORR catalytic performance of Fe₅₀-N-C-900 SAEs in alkaline solution promoted us to study their activity in acid and neutral conditions. Figure 5A shows the RDE polarization curve of Fe₅₀-N-C-900, which possesses onset potential of 0.847 V and high limit current density of 6.83 mA/cm². Based on the RRDE measurement results, the H₂O₂ yield on Fe₅₀-N-C-900 electrode is lower than 10% (Figure 5B), confirming the four-electron pathway. It is worth noting that more than 50% current density was retained on Fe₅₀-N-C-900 electrode after long-term stability test. Whereas, around 70% current loss was observed on Pt/C electrode, as shown in Figure 5C. Remarkably, Fe₅₀-N-C-900 SAEs reveal good ORR catalytic activity in neutral media, which is comparable to commercial Pt/C (Figure 5D). The excellent ORR catalytic performance of Fe₅₀-N-C-900 SAEs over a wide range of pH is closely related to their morphological and compositional advantages. The porous structure with the large surface area can not only provide more exposed active sites but accelerate mass transport and electron transfer. Additionally, the homogeneous single active sites offer high catalytic efficiency.

In summary, Fe-doped MOF-derived Fe-N-C SAEs were successfully synthesized using a facile strategy. The single-atom features were confirmed by XRD, TEM and X-ray adsorption techniques. By virtue of their high surface area and homogeneous single active sites, the Fe-N-C SAEs with optimized synthesis condition possess superior ORR electrocatalytic activity in alkaline media. Remarkably, they present much better durability and methanol tolerance in comparison with the state-of-the-art Pt/C catalysts. Further electrochemical experiments demonstrate that Fe-N-C SAEs also possess excellent ORR catalytic performance in acid

and neutral media. Our work offers a new routine to prepare single-atom catalysts for ORR electrocatalysis.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

This work was supported by a start-up fund of Washington State University, USA. The XPS analysis was performed using EMSL, a national scientific user facility sponsored by the Department of Energy's Office of Biological and Environmental Research and located at Pacific Northwest National Laboratory (PNNL). The authors acknowledge Franceschi Microscopy and Image Center at Washington State University for TEM measurements. PNNL is a multiprogram national laboratory operated for DOE by Battelle under Contract DE-AC05-76RL01830.

Received: ((will be filled in by the editorial staff))

Revised: ((will be filled in by the editorial staff))

Published online: ((will be filled in by the editorial staff))

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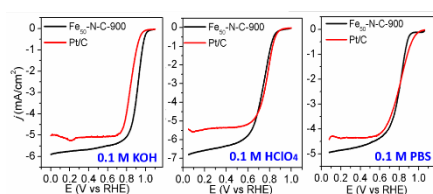
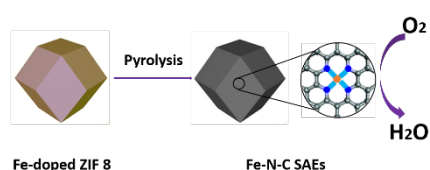
Fe-doped MOF-derived nonprecious nanocatalysts with single active sites were obtained by simple pyrolysis procedure. The obtained Fe-N-C single-atom electrocatalysts possess superior ORR performance in a wide range of pH.

Keyword

Metal-organic frameworks, nonprecious metal nanocatalysts, single-atom catalysts, Fe-N-C electrocatalysts, oxygen reduction reaction

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Porous carbon-hosted atomically dispersed iron-nitrogen moiety as enhanced electrocatalysts for oxygen reduction reaction in a wide range of pH



Supporting Information

Porous carbon-hosted atomically dispersed iron-nitrogen moiety as enhanced electrocatalysts for oxygen reduction reaction in a wide range of pH

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Experimental section

Chemicals and reagents. Iron (III) nitrate nonahydrate (98%, $\text{Fe}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$) and commercial platinum/carbon (Pt/C) 20 wt. % (Pt loading: 20 wt. %, Pt on carbon black) were purchased from Alfa Aesar. Zinc nitrate hexahydrate (98%, $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$), 2-Methylimidazole and Nafion perfluorinated resin solution (5 wt. % in mixture of lower aliphatic alcohols and water, contains 45% water) were obtained from Sigma-Aldrich. Potassium hydroxide (98%, KOH) and methanol (CH_4O) were bought from J. T. Baker. All aqueous solutions were prepared with ultrapure water ($>18 \text{ M}\Omega\cdot\text{cm}$) from Barnstead Nanopure water system.

Apparatus. A FEI Sirion field emission scanning electron microscope (FESEM) was used for imaging. Transmission electron microscopy (TEM) images were obtained by Philips CM200 UT (Field Emission Instruments, USA). X-ray Diffraction (XRD) characterization was carried out by Rigaku Miniflex 600. The tube was operated at 40 kV accelerating voltage and 15 mA current. X-ray photoelectron spectroscopy (XPS) measurements were performed with a Physical Electronics Quantera Scanning X-ray Microprobe. This system uses a focused monochromatic Al $\text{K}\alpha$ X-ray (1486.7 eV) source for excitation and a spherical section analyzer. The electrochemical measurements were carried out by an electrochemical workstation (CHI 630E) coupled with a three-electrode system. A saturated calomel electrode (SCE) filled with saturated KCl aqueous solution and Pt plate was used as reference electrode and counter electrode, respectively.

Prepared of ZIF-8 and Fe-doped ZIF-8 nanocrystals. Typically, 1.68 g of $\text{Zn(NO)}_3 \cdot 6\text{H}_2\text{O}$ was dissolved in 80 mL of methanol to form solution A. 3.7 g of 2-methylimidazole was dissolved in 80 mL of methanol to form solution B. Solution A was then added into solution B with vigorous stirring for 24 hours. The products were then centrifuged and washed for several times before freeze-drying. The preparation of Fe-doped ZIF-8 is same as ZIF-8 except for the addition of certain amount (20, 50 and 100 mg) of $\text{Fe(NO)}_3 \cdot 6\text{H}_2\text{O}$ into solution A.

Synthesis of N-C catalysts and $\text{Fe}_x\text{-N-C}$ SAEs. The above ZIF-8 and Fe-doped ZIF-8 were pyrolyzed under N_2 atmosphere at different temperatures. The final products were denoted as N-C-T and $\text{Fe}_x\text{-N-C-T}$ (x =amount of $\text{Fe(NO)}_3 \cdot 6\text{H}_2\text{O}$, T=temperature).

Electrocatalytic experiments. Prior to the surface coating, glassy carbon rotating disk electrode (RDE, 5 mm in diameter) and rotating ring-disk electrode (RRDE, 6.5 mm in diameter) were polished and rinsed with deionized water, followed by sonicated in ethanol and water. The catalysts (2 mg/mL) were prepared by dissolving the obtained N-C-T, $\text{Fe}_x\text{-N-C-T}$ or Pt/C into Nafion solution which contains Nafion, 2-propanol, and water (V/V/V = 0.025/1/4). The mixture was ultrasonicated to obtain homogeneous catalyst ink. The catalysts (0.1 mg/cm^2) was then dropped on the surface of pretreated RDE/RRDE surface and dried before the electrocatalytic tests.

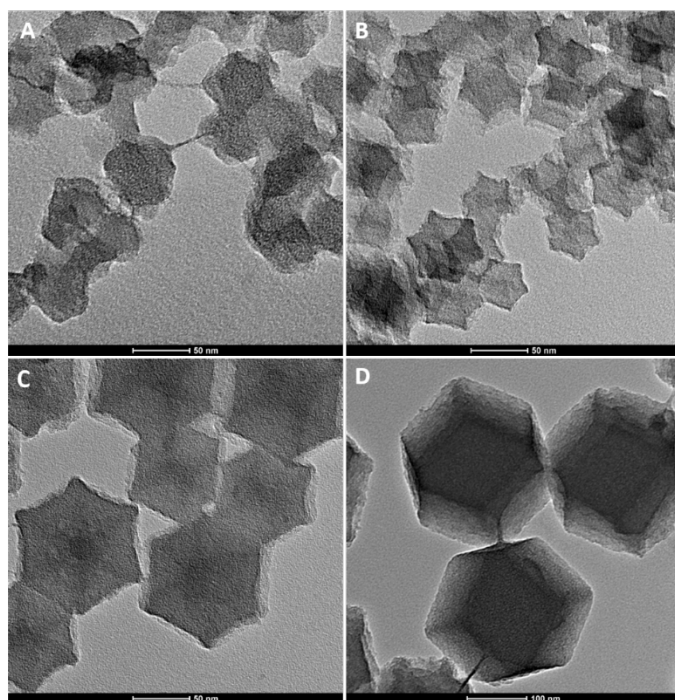


Figure S1. TEM images of N-C-900 (A), Fe₂₀-N-C-900 (B), Fe₅₀-N-C-900 (C) and Fe₁₀₀-N-C-900 (D).

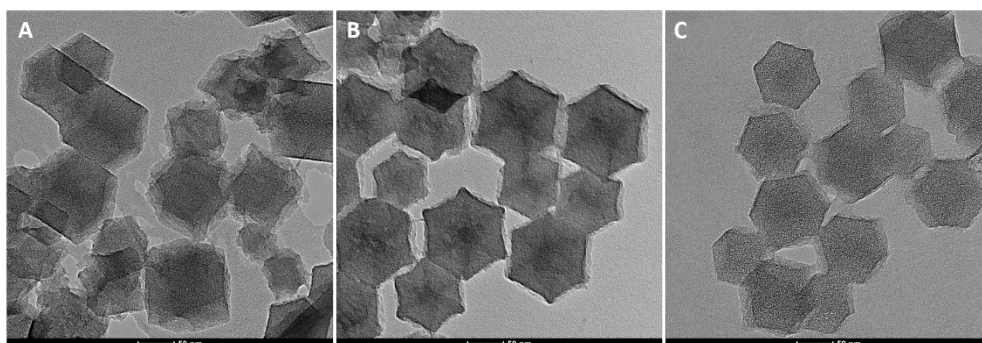


Figure S2. TEM images of Fe₅₀-N-C-800 (A), Fe₅₀-N-C-900 (B) and Fe₅₀-N-C-1000 (C).

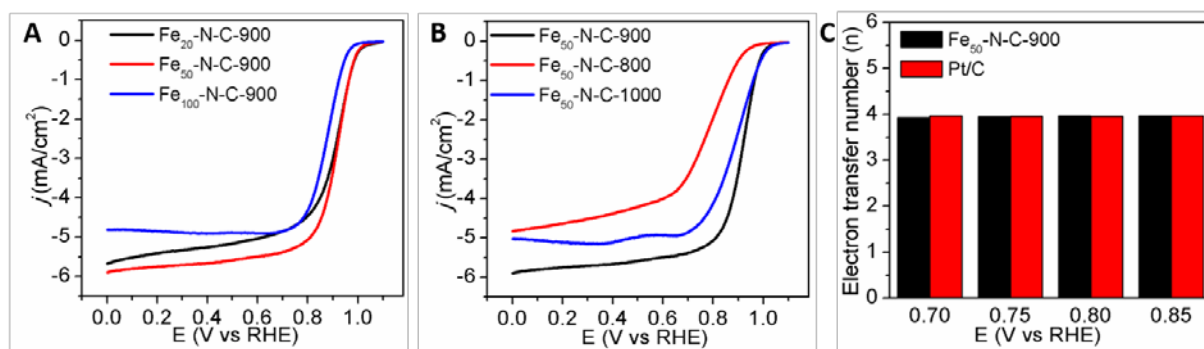


Figure S3. (A) RDE polarization curves of Fe-N-C-900 with different Fe contents. (B) RDE polarization curves of Fe₅₀-N-C with different pyrolysis temperature. The loading is 0.1 mg/cm². (C) Electron transfer number of Fe₅₀-N-C and Pt/C.

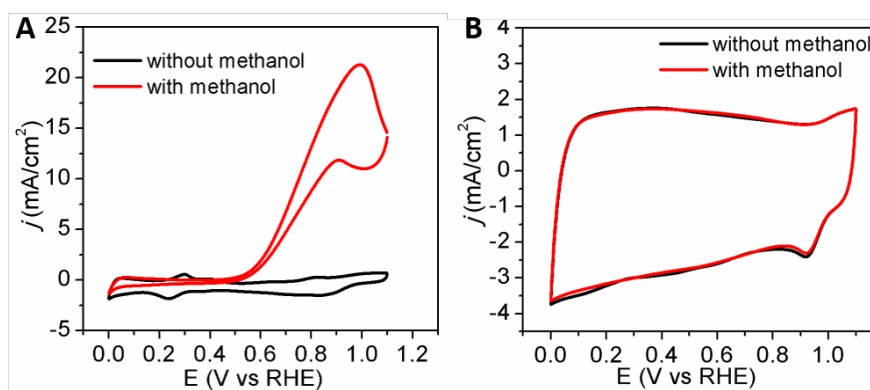


Figure S4. (A) CV curves of Pt/C (A) and Fe₅₀-N-C-900 (B) before and after the addition of 1 M methanol into O₂-saturated 0.1 M KOH solution.

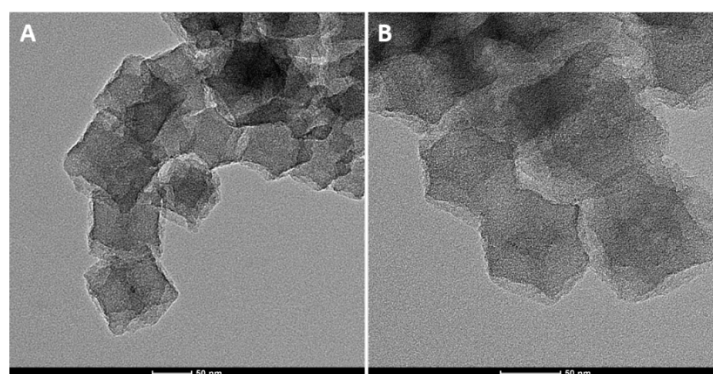


Figure S5. TEM images of Fe₅₀-N-C-900 after stability test.

Table S1. Elemental composition of different electrocatalysts.

Sample	C at%	N at%	O at%	Fe at%	Zn at%
Fe ₅₀ -N-C-800	67.64	17.76	8.71	0.13	5.76
Fe ₅₀ -N-C-900	79.8	10.29	7.06	0.15	2.7
Fe ₅₀ -N-C-1000	84.16	9	4.68	0.14	2.02

Table S2. Comparison of ORR electrocatalytic performances of different materials.

Sample	Medium	rpm	Onset potential (V)	Half-wave potential (V)	Reference
Fe ₅₀ -N-C-900	0.1 M KOH	1600	1.00	0.92	This work
NC900	0.1 M KOH	2500	0.83	NA	J. Am. Chem. Soc. 2014, 136, 6790-6793

Fe-CB@PAN-1000	0.1 M KOH	900	NA	0.758	J. Colloid Interface Sci. 2017, 502, 44-51
5.1Fe-NMC	0.5 M H ₂ SO ₄	1600	0.67	NA	Carbon, 2016, 105, 282-290
(Fe,Fe) ₂	0.5 M H ₂ SO ₄	900	0.835	NA	J. Am. Chem. Soc. 2016, 138, 635-640
ZIF-67-900	0.1 M KOH	1600	0.91	0.85	J. Mater. Chem. A 2014, 2, 14064-14070
ZIF-8-900	0.1 M KOH	1600	0.92	0.84	J. Mater. Chem. A 2014, 2, 14064-14070
CNPs	0.1 M KOH	1600	1.03	NA	ACS Nano 2014, 8, 12660-12668